



Global-local information based oversampling for multi-class imbalanced data

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Abstract

Multi-class imbalanced classification is a challenging problem in the field of machine learning. Many methods have been proposed to deal with it, and oversampling is one of the most popular techniques which alleviates class imbalance by generating instances for the minority classes. However, each oversampling utilizes a single way to generate instances for all candidate minority ones, which neglects the intrinsic characteristics among different minority class instances, and makes the synthetic instances redundant or ineffective. In this work, we propose a global-local-based oversampling method, termed GLOS. We introduce a new discreteness-based metric (DID) and distinguish the minority class from the majority class by comparing it with each class-level discreteness value. Then, for each minority class, some difficult-to-learn instances are selected, which have smaller instance-level dispersion than the corresponding class-level one, to generate synthetic instances. And the number of synthetic instances equals the difference between two types of dispersion values. These selected instances are assigned into different groups according to their local distribution. Furthermore, GLOS adopts a specific synthetic strategy to each group instance purposefully. Finally, all minority classes, part of the majority classes instances, and synthetic data will be used as training data. In this way, the quantity and quality of synthetic instances are guaranteed. Experimental results on KEEL and UCI data sets demonstrate the effectiveness of our proposal.

Keywords Oversampling · Intrinsic characteristics · Synthetic strategy

1 Introduction

Imbalanced data with skewed class distribution exist widely in various applications [1]. In general, some classes may have much more instances than other classes in imbalanced

data sets. Moreover, the ones that have more instances are the majority classes, and the other ones are the minority classes [2]. Although the minority classes have fewer instances than the majority ones, they play essential roles in imbalanced problems. However, the minority class instances are often misclassified by the conventional classification algorithms. In recent years, imbalanced learning has attracted widespread attention, aiming to classify the minority classes correctly and keep the high accuracy of the majority class simultaneously. Many researchers have made extraordinary contributions to dealing with imbalanced problems. However, most of these methods are aimed at two-class imbalanced problems and are not directly applicable to multi-class imbalanced scenarios. The reason is that multi-class imbalanced data has specific characteristics compared with two-class imbalanced data: (a) the relations between classes may be no longer clear, for example, a class will be regarded as the minority class compared to classes having more instances, and conversely, it will be regarded as the majority class with respect to classes having fewer instances; (b) the boundaries between classes are unclear and different

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classes may overlap severely; (c) the minority classes are too unrepresentative to be identified. These factors may make the multi-class imbalanced problem more complicated than the two-class imbalanced one and harder to learn. In addition, Zhou and Liu revealed that almost all techniques are effective on two-class imbalanced problems, which may be ineffective or even cause adverse effects on multi-class tasks when used directly [3]. Therefore, it is meaningful to learn multi-class imbalanced classification problems.

The contributions to handling this problem in the last decade are roughly categorized into two parts: data-level and algorithmic-level. For the former, oversampling [4–6] and undersampling [7–9] are adopted to modify class imbalance. Those have empirically proved that sampling methods effectively balance the class distribution. Furthermore, the main advantage of these techniques is that they are independent of the underlying classifiers and can be implemented easier [2]. The latter mainly includes three categories of algorithms: cost-sensitive learning [10–16], kernel-based methods [17–19] and ensemble learning [20–26]. All of them alleviate class imbalance from the algorithmic view. However, most of them frequently integrate the data level with the algorithmic level method to obtain better results instead of using them alone. Moreover, cost-sensitive methods always rely on a specific cost matrix, which requires deep domain knowledge [27]. Therefore, the data level methods are more commonly used to deal with the imbalanced problem. Moreover, the oversampling method plays the dominant role in handling the imbalance problem because it does not lose vital information.

Oversampling modifies the class imbalance by adding new instances to the minority classes. ROS replicates the original minority class samples and joins them to the corresponding minority class to alleviate the imbalance. Moreover, many oversampling methods such as SMOTE [28], Borderline-SMOTE [29], ADASYN [30], and Safe-Level-SMOTE [31] also can generate new instances in different ways. ROS replicates the minority class instances to balance the class distribution, but it can not increase extra information and may cause overfitting. On the contrary, SMOTE uses an instance and its K neighbors in the same minority class to generate new instances by linear interpolation. Although it neglects the information of the other classes, which may lead to the overlapping of new instances with those of other classes, it is better than ROS. Compared with SMOTE, other methods like Borderline-SMOTE, ADASYN, Safe-Level-SMOTE, and different kinds of SPIDER [27, 32] generate new instances by considering the local distribution of each instance differently. Thus, they can improve synthetic instances' quality and reduce overlap to a certain extent. Some approaches take the overall distribution of classes into account to generate high-quality instances such as MDO [33], AMDO [34], and SWIM [35]. By sufficiently considering the distribution of individual

classes and utilizing a covariance matrix to generate instances, the synthetic instances can keep the minority class structure initially as much as possible. MC-BRO [36] takes into account information of all classes and modifies the parameters according to our intentions to generate instances, which can make a trade-off between precision and recall. From a new perspective, Li [37] proposed three sampling methods based on a new imbalance metric (EID). Furthermore, these methods balance a data set after evaluating the information influence degree of each instance. For example, EOS generates new instances near the informative ones, EUS removes easy-to-learn instances, and EHS integrates EUS and EOS. However, for each oversampling method, all synthetic instances are generated using a single strategy, which may overlook the intrinsic characteristics of instances and generate useless synthetic instances. As such, the synthetic instance may not effectively increase the representativeness of the corresponding minority class and may even deteriorate the accuracy of the majority class.

In this work, we propose a global-local based oversampling method, in which three synthetic strategies are designed for the minority class instances having various intrinsic characteristics. The first one takes the covariance and mean of all majority classes to generate instances for the isolated minority class instances. The synthetic instances have the same distance with the reference instances to different majority class means. The second one considers the distribution of specific instance's density area within global data and generates instances adaptively by liner-interpolation to shift the boundaries toward the majority classes. The third one only considers the distribution of each sub-cluster and utilizes radial-based random oversampling to generate instances to expand their local space. Moreover, we also propose a new metric, DID, which provides a clear standard to distinguish the majority and minority classes and adaptively determines the number of synthetic instances for each minority instance. Doing so may avoid generating too many redundant instances to cause overfitting and strengthen the quality of synthetic instances as much as possible.

The rest of the paper is organized as follows. Section 2 presents the pipeline of GLOS and provides a demo to show how GLOS works. Our proposed approach will be described in Sect. 3. In Sect. 4, the experimental results and analyzes are presented to explore the effectiveness and robustness of GLOS. Section 5 makes a summary for the paper and has the outlook of our future work.

2 Overview of the proposal GLOS

This section briefly introduces our proposal GLOS and provides a demo to show how it works intuitively. Figure 1 shows the pipeline of GLOS, and it mainly includes three parts: instance selection, instance partition, and different oversampling strategies.

Figure 2 shows decision boundaries on different phases of GLOS. In Fig. 2, the blue points represent the majority class, the rest are the minority classes, and the shaded area covered by the same color as the points represents the decision area of the corresponding class. Figure 2a shows the decision boundary on the original data, and it can be observed that all yellow points can not be recognized. Figure 2b shows the change of decision boundary after two synthetic strategies oversample the minority classes.

Compared with Fig. 2a, the most yellow points are identified correctly, and the purple shaded area becomes more precise in Fig. 2b. Figure 2c shows the decision boundary on data sampled by GLOS thoroughly. Although the purple shaded area covers a few yellow points, it covers all purple points well. Moreover, the yellow shaded area also covers all the yellow rest points, which are not covered in Fig. 2b. Compared with the former two cases, the more minority class instances are recognized correctly in Fig. 2c.

Fig. 1 The pipeline of GLOS

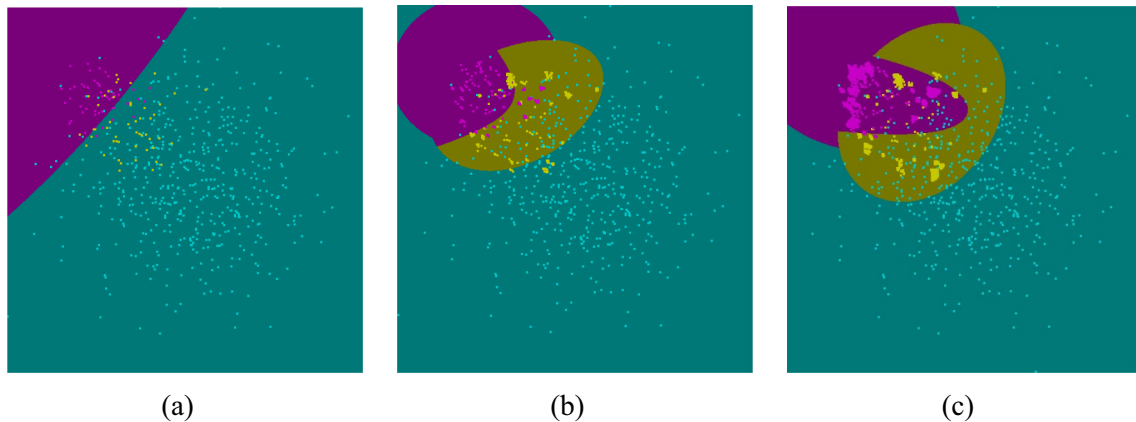
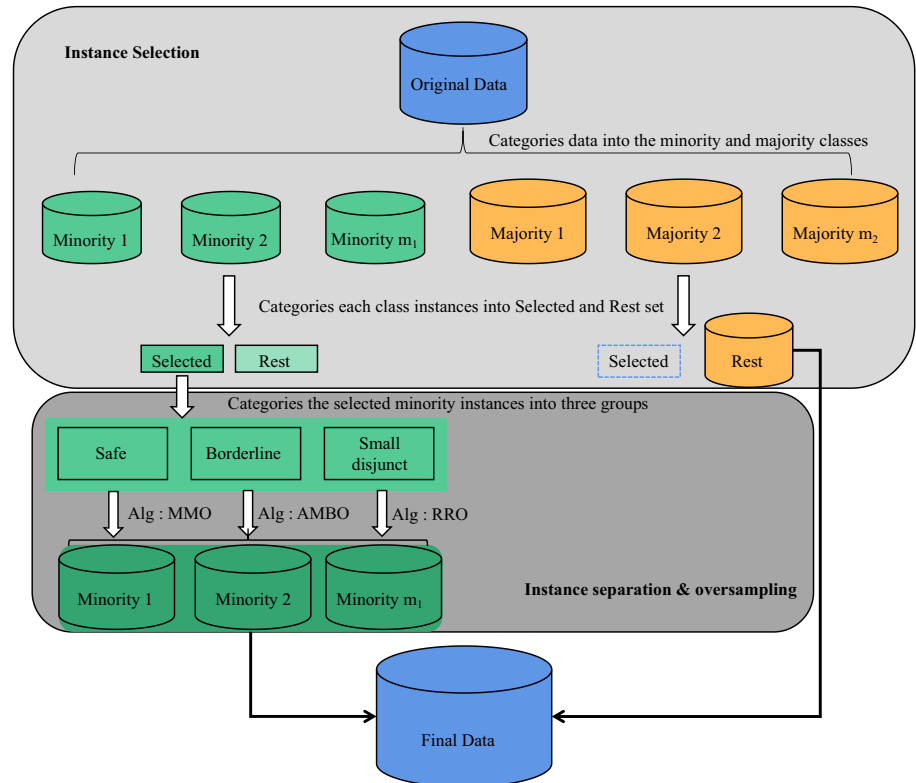


Fig. 2 Decision boundaries changes with GLOS

3 The proposed method

3.1 DID: Discreteness-based imbalance degree

The instance discreteness may severely affect the learning ability of instances. As an imbalance metric, IR only describes the information about the size of instances between the most majority class and the most minority class and fails to evaluate it. Fortunately, the standard deviation can measure the degree of the data dispersion and provide information on the tightness and consistency of the data as well. Therefore, we first measure the instance-level dispersion and reflect the class-level dispersion by averaging the instance-level dispersion of all instances within the corresponding class. Moreover, the new proposed measure, DID, equals the weighted class-level dispersion to evaluate the overall discreteness of dataset.

Given a data set D that includes N instances $X = \{x_1, x_2, \dots, x_N\}$ with $x_i \in \mathbb{R}^d$, where there are m classes $C = \{c_1, c_2, \dots, c_m\}$ with corresponding number of instances of each class $\{C_1, C_2, \dots, C_m\}$ and $N = C_1 + C_2 + \dots + C_m$. DID , is equal to the weighted average class-level discreteness, and is defined as (1).

$$DID = \sum_{j=1}^m \theta_j \frac{C_j}{N}. \quad (1)$$

Where θ_j measures the average dispersion of instances within class j and can be calculated according to (2). Specifically, the lower θ_j is, the sparser the corresponding class and the more likely it is difficult to learn. Moreover, we define it as the minority class.

$$\theta_j = \frac{1}{C_j} \sum_{i=1}^{C_j} \lambda(x_i). \quad (2)$$

Where $x_i \in c_j$, and $\lambda(x_i)$ measures the discreteness of x_i and is shown as (3). For a instance x_i , the smaller $\lambda(x_i)$ is, the more likely x_i located in a sparser region and more difficult to learn. Here, we set the size of synthetic instance of x_i as the difference between $\lambda(x_i)$ and the corresponding θ_j .

$$\lambda(x_i) = |\{x_{in} \mid \|x_{in} - x_i\|_2 \leq R, x_{in} \in c_j \wedge x_i \in c_j\}|. \quad (3)$$

Where x_{in} belongs to the same class as x_i and $\lambda(x_i)$ equals the number of instances located in specific region, which is a hypersphere with x_i as the center and R as radius. Exactly, R is defined as (4),

$$R = \alpha \cdot \min\{sd_1, sd_2, sd_3, \dots, sd_d\}, \quad (4)$$

where sd_k is the standard deviation of the k -th feature of D and α is a hyper-parameter in tuning the radius, and we set $\alpha \in \left(1, \frac{\max(sd)}{\min(sd)}\right)$ and $sd = \{sd_1, sd_2, \dots, sd_d\}$.

3.2 Instance selection

We categorized data into the majority and minority classes firstly,

$$\begin{cases} \text{Majority} = \{c_i \mid \theta_i \geq DID\} \\ \text{Minority} = \{c_i \mid \theta_i < DID\}. \end{cases} \quad (5)$$

Then, instances within each class will be assigned to *Selected* and *Rest* as (6),

$$\begin{cases} \text{Selected}_j = \{x_i \in c_j \mid \lambda(x_i) < \theta_j\} \\ \text{Rest}_j = \{x_i \in c_j \mid \lambda(x_i) \geq \theta_j\}. \end{cases} \quad (6)$$

And, the minority instances in *Selected* are regarded as primary instances for data generation, and all instances in *Rest* are regarded as assistant ones to participate in different synthetic approaches to generate new instances.

3.3 Instances partition

Though underrepresented in their classes, the selected instances have different intrinsic characteristics, mainly referring to locating in small disjuncts, deviating from most instances, or distributing in overlapped regions. Using the same way to generate instances may overlook their distinctive intrinsic characteristics. Therefore, we assign these instances into three groups and propose different synthetic strategies for each group. Moreover, the selected majority instances will be removed as they may cause extra overlap and make the minority class hard to learn. Here, we divided the minority class instances into three groups: *Safe*, *Borderline*, and *Small Disjunct*.

Safe: This kind of instance far away from others and may increase the diversity of its corresponding class. Here, we defined it as follows:

$$\begin{aligned} \text{Safe} = \{x_i \mid x_i \in \text{Minority} \wedge \lambda^D(x_i) \\ - \lambda(x_i) = 0 \wedge \lambda(x_i) \leq 5\}, \end{aligned} \quad (7)$$

and $\lambda^D(x_i)$ is different from $\lambda(x_i)$ that it includes samples from different classes.

$$\lambda^D(x_i) = |\{x_{in} \mid \|x_{in} - x_i\|_2 \leq R, x_{in} \in D\}|. \quad (8)$$

Borderline: This type of instance may locate in class boundaries and near to different class samples. In classification tasks, these instances significantly affect classification performance, especially for the complicated multi-class imbalanced task. Here, we defined it as follows:

$$\text{Borderline} = \left\{x_i \mid 0.5 \leq \frac{\lambda(x_i)}{\lambda^D(x_i) - \lambda(x_i)} \leq 2\right\}. \quad (9)$$

Here, 0.5 and 2 are empirical settings that applied in all experiments in this work.

Small Disjunct: This type of sample splits the corresponding class into different sub-clusters, resulting in the class being underrepresented. They may be located inside other classes or in sub-clusters of their corresponding class, as defined below:

$$\text{SmallDisjunct} = \{x_i \mid x_i \in c_j \wedge x_i \in \{\text{Selected}_j - \text{Safe} - \text{Borderline}\}\}. \quad (10)$$

Figure 3 shows the distribution of different types of instances within an imbalanced dataset to more intuitively show the changes of the dataset after instance selection and partition.

3.4 Oversampling for instances in different groups

GLOS distinguishes the intrinsic characteristics of different types of instances and uses different sampling strategies to purposely generate instances that will enlarge the minority class space and make the minority classes more representative.

Safe samples are located in sparse regions or far away from other samples, so they are not closely related to other samples. SMOTE-based methods can expand the distribution area of these samples, but it increases the risk of overgeneration. Gaussian noise-based methods can avoid overgeneration, but it relies on the parameters of the Gaussian distribution and is prone to overfitting. MDO-based methods can preserve their position information and increase their diversity, but the covariance matrix is not easy to obtain. Inspired by MDO, we propose the MMO strategy for **Safe** samples. It uses the covariance matrix of the majority class for data generation and limits the regions of the synthesized samples, which reduces the risk of overgeneration to a certain extent.

For each sample, the MMO first defines the maximum region of the synthetic sample according to Eq. 8. Then calculates the covariance matrixes of the majority classes and generates samples, respectively. These samples must be located in the maximum space and have the same Mahalanobis distance to the center of the majority classes as the corresponding **Safe** sample. Algorithm 1 shows the detailed process of MMO.

Algorithm 1 MMO: Majority-based Minority Oversampling

Require: The reference instance x_r , $\lambda(x_r)$

Ensure: The synthetic instances set X_{syn}

```

1:  $syn_{num} \leftarrow DID - \lambda(x_r)$ 
2: while  $syn_{num} > 0$  do
3:   for  $j = 1 : m$  do
4:     acquire  $mean_j$  and  $\Sigma_j$ 
5:      $dist_{x_r}^{mean_j} \leftarrow \sqrt{(x_r - mean_j)^T \Sigma_j^{-1} (x_r - mean_j)}$ 
6:     acquire sphere built by  $x_r$  and  $R$  using (3)-(4)
7:     generate  $x_{new}$  with conditions that:
8:     1.  $dist_{x_r}^{mean_j} \leftarrow \sqrt{(x_{new} - mean_j)^T \Sigma_j^{-1} (x_{new} - mean_j)}$ 
9:     2.  $|x_{new} - x_r| \leq R$  (calculated in line 6)
10:     $X_{syn} \leftarrow X_{syn} \cup x_{new}$ 
11:     $syn_{num} \leftarrow syn_{num} - 1$ 
12:   end for
13: end while
```

Figure 4 shows the distribution of synthetic samples generated by MMO. The black points are synthetic instances, and the black circle is the maximum region that synthetic samples located in. The black points located in blue, red, and green curves have the same Mahalanobis

distance as the primary sample to the blue, red, and green classes, respectively. Moreover, these synthetic instances may keep the minority classes structures initially and strengthen the representativeness of **Safe** instances.

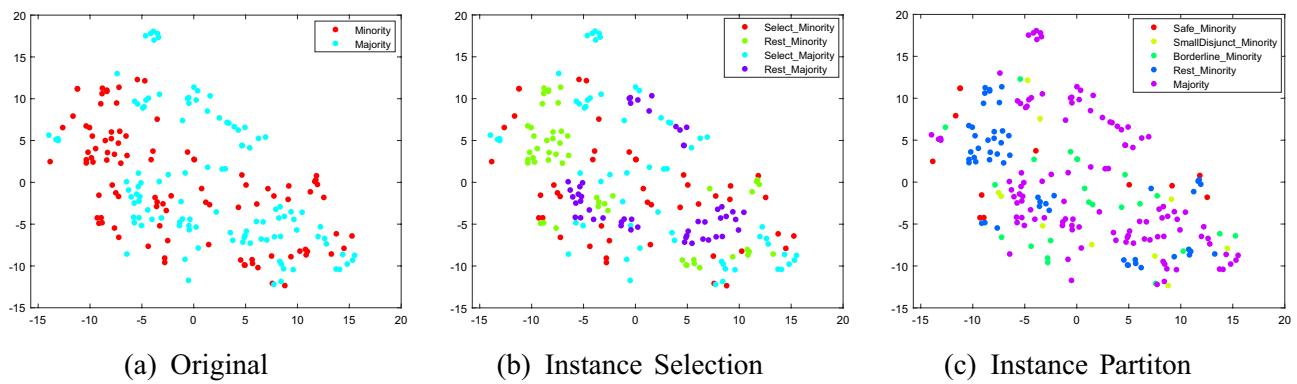


Fig. 3 Distribution of different types of samples in imbalanced dataset

Algorithm 2 AMBO: Adaptive Multi-class Borderline Oversampling

Require: The reference instance x_r , $\lambda(x_r)$

Ensure: The synthetic instances set X_{syn}

- 1: $syn_{num} \leftarrow DID - \lambda(x_r)$
 - 2: acquire the sphere built by x_r and R using (3)-(4)
 - 3: calculate num_c in the sphere defined as line 2
 - 4: **if** $c \in Majority$ **then**
 - 5: $num_c \leftarrow num_c * \frac{\theta_c}{\theta_i}$
 - 6: **else**
 - 7: $num_c \leftarrow num_c * \frac{\theta_i}{\theta_c}$
 - 8: **end if**
 - 9: **normalization** num_c
 - 10: $D(x_r) \leftarrow \min\{dist(x_r, x_i)\}$, where $x_i \in X$ and X is defined by line 2
 - 11: $syn_c \leftarrow syn_{num} * num_c$
 - 12: $x_{new} \leftarrow x_r + (x - x_r) * \frac{D(x_r)}{[(x - x_r)]} * rand(0, 1)$
 - 13: $X_{syn} \leftarrow X_{syn} \cup x_{new}$
-

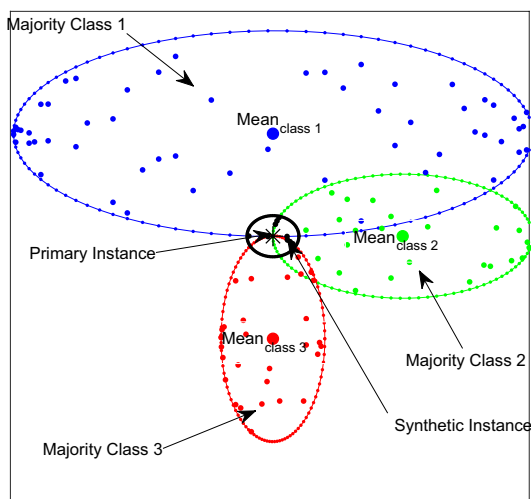


Fig. 4 Distribution of synthetic samples generated by MMO

Borderline samples are located in the decision boundary or the class overlap area and are often closely related to samples of different classes, but it is also the main reason for the difficulty of sample learning, especially for the minority class samples. In multi-classification tasks, as the number of classes increases, the complexity of samples near boundaries also increases, and how to enhance the representativeness of minority classes is more important. The samples generated by the SMOTE-based method can strengthen the representation ability of such samples near boundaries so that the decision boundary moves to the majority class, thereby improving the learning ability of the minority class. However, when faced with multi-classification tasks, such methods make other minority classes harder to learn. Therefore, we propose AMBO to generate new samples for **Borderline** cases, as shown in Algorithm 2.

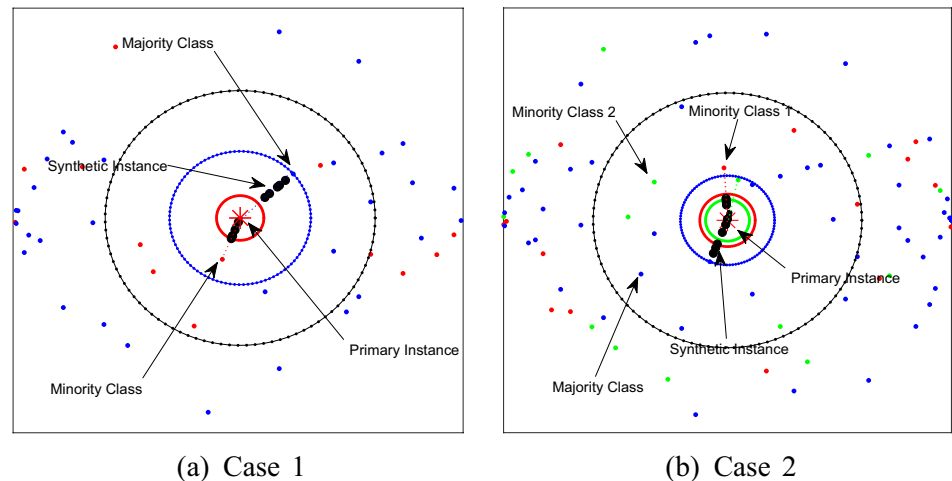
Fig. 5 Distribution of synthetic samples generated by AMBO

Figure 5 shows the instance distribution processed by AMBO, which considers two kinds of **Borderline** instances distribution. One has only two class instances, and the other includes multiple minority and majority classes. In both cases, the samples generated by AMBO are closer to the majority class to ensure that the boundary moves towards the majority class. In the case of containing other minority classes, the instance (black dot) synthesized by the reference sample and other minority class samples are closer to the reference sample to keep the structure initially between the minority classes as much as possible to avoid overgeneration on other minority classes.

Small Disjunct samples often appear in subclusters, which is one of the main factors that make decision

boundaries complex and samples difficult to learn. Effectively expanding the distribution area of these samples is of great significance for improving the learning performance of the minority class. The SMOTE-based method will synthesize redundant samples and lead to overfitting, and the linear interpolation method is not conducive to expanding the potential distribution area of these samples. Similarly, AMBO has the above problems when synthesizing data for such samples. This paper uses RRO to generate instances for **Small Disjunct** samples to deal with the above problem. It expands the instance space and significantly reduces the generation of redundant samples. See Algorithm 3 for details.

Algorithm 3 RRO:Radial-based Random Oversampling

Require: The reference instance x_r , $\lambda(x_r)$

Ensure: The synthetic instances set X_{syn}

- 1: $syn_{num} \leftarrow DID - \lambda(x_r)$
 - 2: obtain the sphere built by x_r and R using (3)-(4)
 - 3: calculate the number of classes $count$ in line 2
 - 4: **if** $count = 1$ **then**
 - 5: calculate the average distance D_{avg} between x_r and other instances
 - 6: generate instance x_{new} with condition that $|x_r - x_{new}| \leq D_{avg}$
 - 7: **else**
 - 8: $mean_r \leftarrow \frac{x}{m}$, $target(x) = target(x_r)$
 - 9: $D_{max} \leftarrow \max(|mean_r - x|, target(x) \neq target(x_r))$
 - 10: generate instance x_{new} with condition that $|mean_r - x_{new}| \leq D_{max}$
 - 11: **end if**
 - 12: generate syn_{num} instances to X_{syn}
-

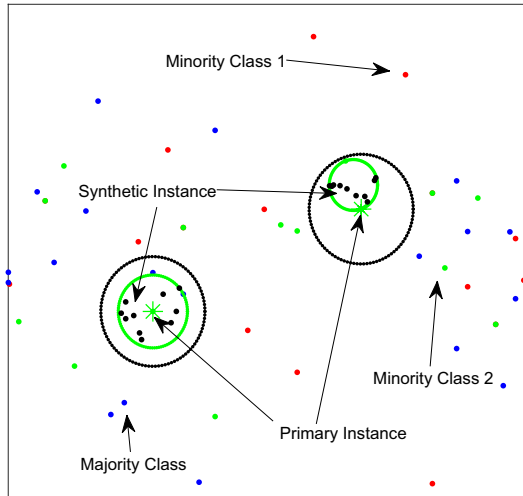


Fig. 6 Distribution of synthetic samples generated by RRO

Figure 6 shows how RRO generates instances and their corresponding distributions for **Small Disjunct** samples through two examples. And Algorithm 4 shows the full implementation of our method in detail.

3.5 Computational complexity analysis

We revisited the pipeline of GLOS, which mainly involves four main modules: class division, instance selection, instance partition, and data generation. The remaining three modules must be executed sequentially except for the data generation module. The complexity of calculating DID and class division is $\mathcal{O}(N_{max})$, N_{max} is the number of instances in the largest class. And the complexity of instance selection is $\mathcal{O}(N_{max})$. Instance partition needs to consider the global distribution of samples, and its computational complexity is $\mathcal{O}(N_{Selected}N)$, $N_{Selected}$ is the number of instances used for data generation and N is the total number of instances of the overall dataset. For different oversampling methods, we assume that the number of samples synthesized by each method and used for data generation are identical and equal to $N_{Primary}$ and N_{syn} . The complexity of RRO is $\mathcal{O}(N_{Primary}N + N_{syn}d)$, and the complexity of AMBO is $\mathcal{O}(N_{Primary}N + N_{syn}d)$. MMO requires matrix decomposition, which is computationally costly, and its computational complexity is $\mathcal{O}(N_{max}d^2 + N_{syn}d)$, where d is the number of features. To sum up, the computational complexity of our proposal is $\max(\mathcal{O}(N_{max}d^2), N_{Selected}N)$, which is approximately equal to $\mathcal{O}(N^2)$.

Algorithm 4 GLOS: Global-Local based Oversampling

Require: Training data X and α

Ensure: The final Training Set X_{new}

```

1:  $\theta \leftarrow \{\theta_1, \theta_2, \dots, \theta_m\}$  using (2)
2: acquire  $DID$  using (1)
3: for  $j = 1 : m$  do
4:   for  $i = 1 : C_j$  do
5:     if  $\lambda(x_i^{C_j}) \leq \theta_j$  then
6:        $Selected_j \leftarrow Selected_j \cup x_i^{C_j}$ 
7:       if  $\theta_j \leq DID$  then
8:         categories  $x_i^{C_j}$  into specific group  $G$ 
9:         Using specific algorithm to generate samples according to  $G$ 
10:      else
11:        removes  $x_i^{C_j}$ 
12:      end if
13:    else
14:       $Rest_j \leftarrow Rest_j \cup x_i^{C_j}$ 
15:    end if
16:  end for
17:   $X_{new} \leftarrow X_{new} \cup X_{syn}$ 
18: end for
```

Table 1 Description of 21 UCI and KEEL Benchmark Datasets with Their Characteristics

ID	Dataset	#Class	#Size	#Features	IR
1	Balance	3	625	4	5.96
2	Breast-tissue	6	106	9	1.63
3	Cardiotocography	3	2126	35	9.40
4	Cleveland	5	297	13	12.00
5	Dermatology	6	366	35	5.58
6	Glass	6	214	9	9.20
7	Hayesroth	3	132	5	1.72
8	Pageblock	5	5473	10	173.40
9	User Knowledge Modeling(UKM)	4	258	5	3.79
10	Wine	3	178	13	1.48
11	Winequality-red	6	1599	12	68.2
12	Winequality-white	5	4888	11	13.8
13	GesturePhaseSegmentationProcessed	5	9873	32	3.00
14	Ecoli	5	327	7	7.17
15	Yeast	9	1479	8	23.2
16	Segment	7	2310	18	9.9
17	Segment_2	7	528	18	9.9
18	Segment_6	7	2013	18	9.9
19	Texture	11	2800	40	10
20	Texture_s	11	1000	40	10
21	Texture_l	11	5050	40	10

4 Experiment results and analysis

4.1 Data sets and experimental setting

21 multi-class data sets from the UCI repository and KEEL repository [38] are selected to evaluate the effectiveness of our proposed algorithm. Firstly, we removed the instances with missing values from the data sets. Secondly, to meet the experiment's requirements, we removed the classes with less than 10 instances in each data set and adjusted the size of classes in some data sets. Table 1 shows the characteristics of the above data sets in detail.

We compare GLOS to 5 state-of-the-art methods like SMOTE, Borderline-SMOTE, ADASYN, MDO, and MC-BRO introduced in Sect. 1 comprehensively. We set K nearest neighbors to 5 in all methods concerning the parameter settings, which is the most accepted value in literature. For ADASYN, $\beta = 1$, as suggested in [30]. As the authors advised, $K1$ and $K2$ are equal to 5 and 10 in MDO, which are the most acceptable settings. After some preliminary experiments, we set $\alpha = 2$ empirically in GLOS, as it achieves relatively good results with this setting. The C4.5 with default parameters is used as the base classifier in all experiments, as it is insensitive to imbalanced data. We conducted five-fold cross-validation with ten independent runs for our experiments. Except for MC-BRO and GLOS that determine the number of synthetic instances by themselves, the

number of instances generated by other methods is equal to $N_{max} - N_i$. Moreover, N_{max} is the number of instances within the most majority class, and N_i is the number of instances within class i . We utilize four class decomposition strategies such as OAA, OAO, OAHO, and AandO to combat multi-class classification problems in MATLAB versions [39]. Furthermore, we selected the best result of all decomposition strategies as the final results of each method to analyze the performance of all approaches.

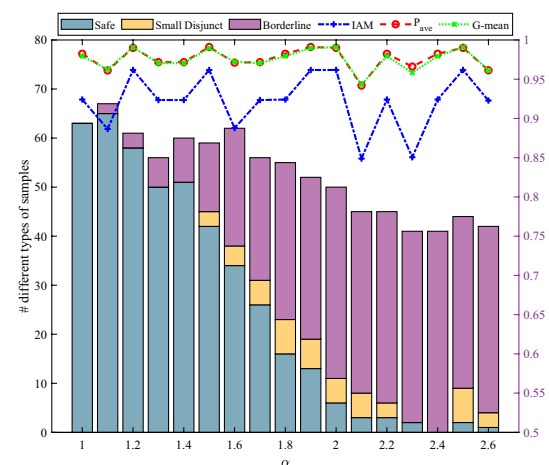


Fig. 7 The impact of α on the results of instance partition and classification performance of GLOS

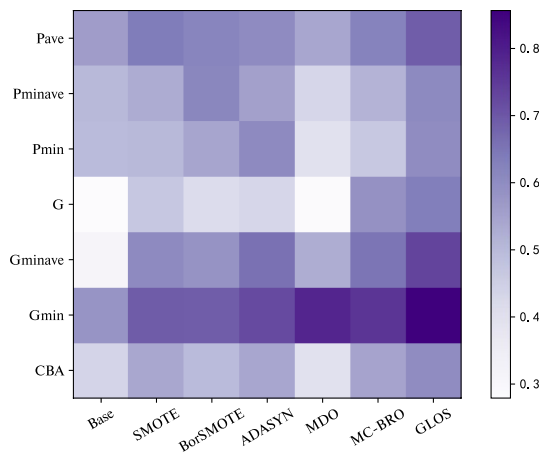


Fig. 8 The mean results of all methods on different metrics (datasets 1–15)

4.2 Evaluation metrics

Traditionally, the most widely used metrics for classification are accuracy and error rate. However, they can be deceiving in imbalanced situations [40]. Therefore, we take the overall performance, the performance of all minority classes, and the most minor class performance into consideration to evaluate the effectiveness of all

methods. Especially, recall R and precision P are used to measure the results of each component.

$$P_i = \frac{TP_i}{TP_i + FP_i}. \quad (11)$$

$$R_i = \frac{TP_i}{TP_i + FN_i}. \quad (12)$$

Where TP_i is the number of well-classified instances within class i and FP is the number of instances classified as class i incorrectly, FN_i is the number of instances classified incorrectly within class i . For the smallest minority class, P_{min} and G_{min} are calculated according to P and R , respectively. And we modified $G - mean$ [41, 42] as G and introduced P_{ave} [43–45] to evaluate the multi-class overall performance as follows:

$$G = \sqrt[c]{\prod_{i=1}^c R_i}, \quad (13)$$

$$P_{ave} = \frac{1}{c} \sum_{i=1}^c P_i, \quad (14)$$

where c is the number of classes. And we also defined P_{minave} and G_{minave} to measure the overall performance of all minority classes, as follow:

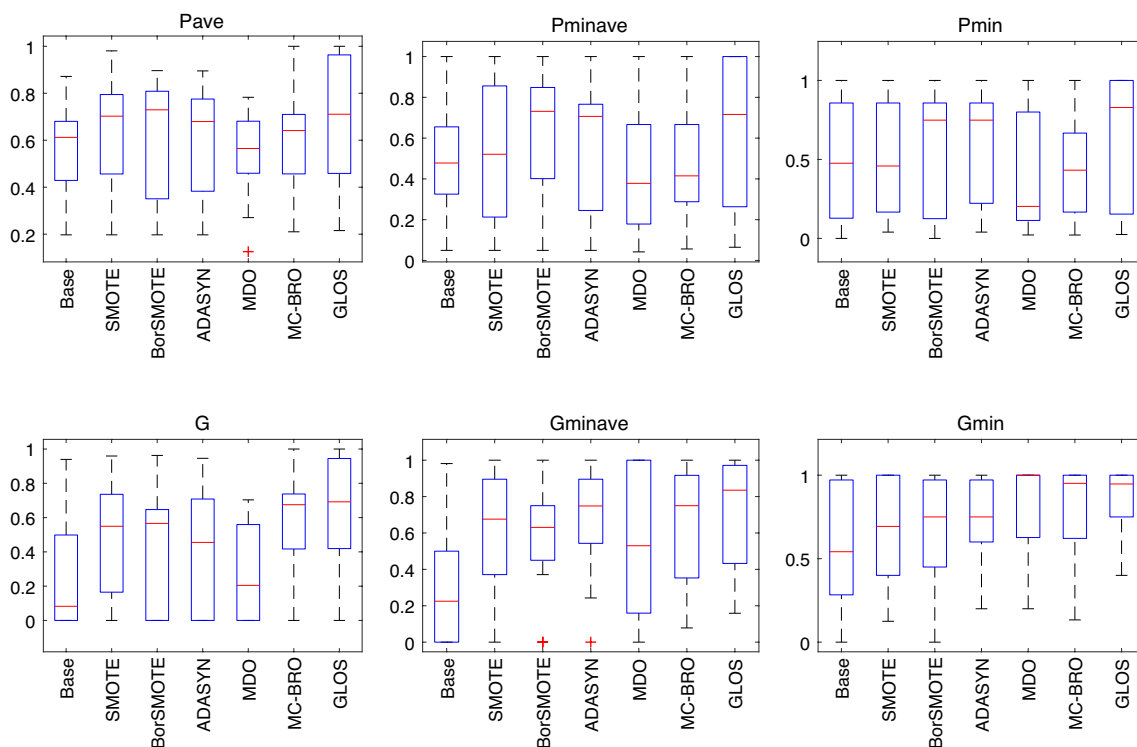


Fig. 9 The results of all methods on different metrics (datasets 1–15)

$$P_{minave} = \frac{1}{c_{min}} \sum_{i=1}^{c_{min}} P_i, \quad (15)$$

$$G_{minave} = \frac{c_{maj}}{\sqrt{\prod_{i=1}^{c_{min}} R_i}}, \quad (16)$$

where c_{min} is the number of the minority classes.

Moreover, we also use *CBA* to replace accuracy, which is a skew-insensitive class balanced accuracy. As shown in (17), mat_{ij} stands for the number of instances of the true class i , which were predicted as class j . For all mentioned metrics, the higher the value, the better the classification outcome.

$$CBA = \frac{\sum_{i=1}^c \sum_{j=1}^c \frac{mat_{ij}}{\max(\sum_{i=1}^c mat_{ij}, \sum_{j=1}^c mat_{ij})}}{c}. \quad (17)$$

4.3 Observations and analysis

In this subsection experiments are performed with two purposes:

1. Comparing GLOS with other oversampling techniques to examine the effectiveness of GLOS on numeric imbalanced data sets;
2. Examining the robustness of GLOS on different kinds of imbalanced data sets: multi-minority, multi-majority, and the general case (multi-minority along with multi-majority).

Figure 7 takes the benchmark dataset Sonar as an example to show the number of different types of samples after instance partition with different α and the changes of classification results on dataset processed by GLOS on specific α , respectively. We observe that as α increases the number of samples assigned to **Safe** decreases and the number of **Small**

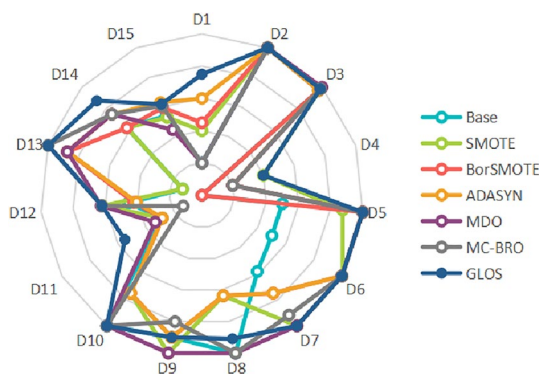


Fig. 10 The (*CBA*) results of all methods (datasets 1-15)

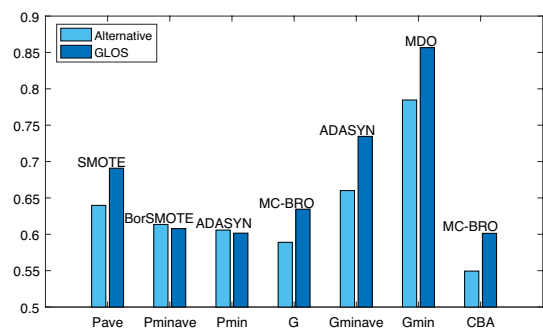


Fig. 11 The Top2 results on all metrics (datasets 1–15)

disjunct and **Boederline** samples increases dramatically, especially for **Boederline** samples. Moreover, three classification metrics changed with α , which are optimal at $\alpha = 2$. In order to synthesize diverse and effective samples, it is necessary to utilize different oversampling methods, to generate samples, and $\alpha = 2$ in our experiments settings. However, it is not necessarily the best value for all datasets. Additionally, we recommend setting $\alpha \in \left[1, \frac{\max(sd)}{\min(sd)}\right]$.

Figure 8 shows the mean results of all methods on different metrics, and the darker the color of the block, the better the corresponding result. In this picture, the color of blocks in the last column is darker than the blocks of other columns on each row, especially in P_{ave} , G , G_{minave} and G_{min} rows. It indicates that GLOS is more effective than other methods in recalling different class instances, especially for the minority classes. It can not deteriorate the majority class accuracy severely compared with other methods. The color of blocks of GLOS in P_{minave} and P_{min} rows are darker lightly than most of the other columns except BorSMOTE and ADASYN,

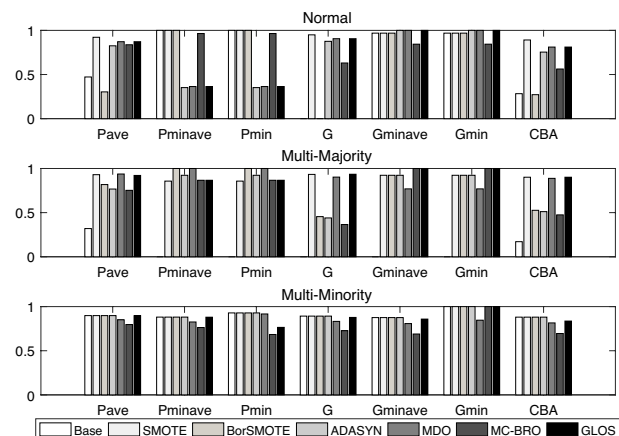


Fig. 12 The results of all methods on different metrics in different kinds of imbalanced cases (*Segment*)

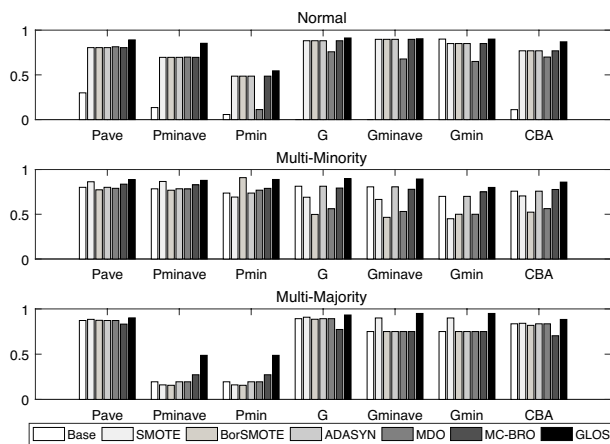


Fig. 13 The results of all methods on different metrics in different kinds of imbalanced cases (*Texture*)

which indicates that BorSMOTE and ADASYN make the better prediction of the minority classes than GLOS and other methods. However, the color of other blocks of GLOS is darker than ADASYN and BorSMOTE. Besides the first column, the color of blocks in the MDO column are paler than others for each row except G_{min} . This may be because MDO focuses more on the minority classes and cause severe over-generation that deteriorate the majority classes. Besides MDO, the base classifier also obtains $G = 0$, compared with other methods. It tells us that the right oversampling is necessary for improving the minority class performance in imbalanced learning.

Figure 9 shows the results of methods on different metrics. From the result of P_{ave} , it can be seen that almost all methods have better P_{ave} than the base one, except MDO and MC-BRO, which have the same results as the base cases. Though the boxes of BorSMOTE, ADASYN, and GLOS are wider, which means they may have unstable performance in various data sets, GLOS has relatively better overall results

than the two others. From P_{minave} , we can see that the lower quartiles of some methods such as SMOTE, MDO, and ADASYN are smaller than the base ones. This is because the generated instances may cause overlaps, and more majority classes instances are classified as the minority classes. However, SMOTE and ADASYN have better overall results than MDO, as MDO may cause more severe overgeneration than two others. GLOS obtains the best results among all methods, as the median of GLOS is larger than the upper quartiles of others, and GLOS obtains $P_{minave} = 1$ in part of data sets. For the most minority class, GLOS also gets the best results. However, MDO and MC-BRO are worse than the base results. As we can see from G , G_{minave} and G_{min} , all methods have better results than the baseline, which indicates that more the minority classes instances are identified after the data is processed by various oversampling. The data processed by MDO, MC-BRO, and GLOS obtain the better G_{minave} and G_{min} than other methods. However, compared with MC-BRO and GLOS, MDO generates instances that severely deteriorate the majority classes' performance, as $G=0$ can be seen from subfigure G . ADASYN and BorSMOTE also have the same results as MDO because they may focus on the minority classes and cause overgeneration. Considering all metrics, GLOS gets better performance than other methods, which indicates that GLOS can be the first choice to handle multi-class imbalanced problems directly. Moreover, SMOTE is better than other methods except for GLOS, which can also simultaneously obtain better recall and prediction.

Figure 10 shows the CBA result of all methods in data sets 1–15. In this figure, the closer the point is to the center, the worse the result. As we can see, GLOS obtain the best results in most data sets except data set 8–9, which means GLOS has better overall performance than other methods. Besides the base classifier, three methods such as SMOTE, BorSMOTE, and MC-BRO also receive the worst results in

Fig. 14 The mean results of all methods on different metrics (datasets 16–21)

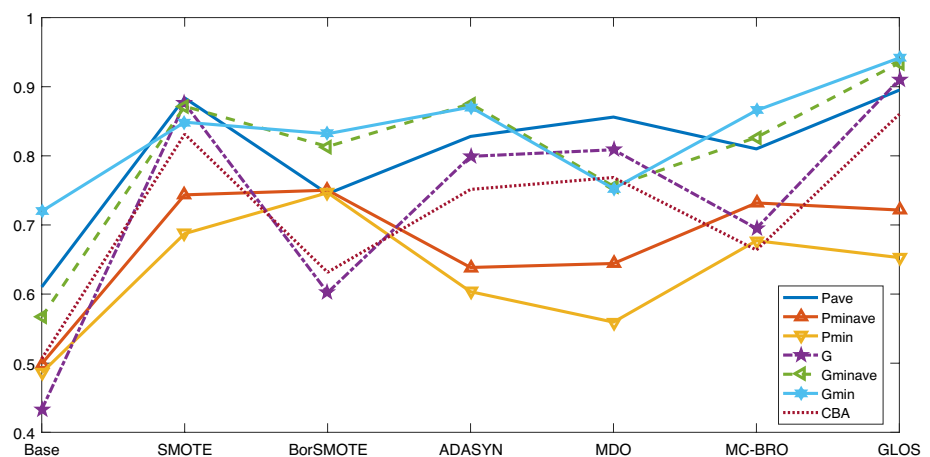


Table 2 The Wilcoxon test for GLOS and comparative algorithms on Recall

VS		R^+	R^-	Exact P -value	Asymp- totic P -value
G_{min}	Base	85.5	5.5	0.0029	0.0042
	SMOTE	80.0	11.0	0.0134	0.0136
	BorSMOTE	100.0	5.0	0.0012	0.0019
	ADASYN	100.0	5.0	0.0012	0.0022
	MDO	60.0	31.0	≥ 0.2	0.2902
	MC-BRO	72.5	18.5	0.0628	0.0546
G_{minave}	Base	88.0	3.0	0.0012	0.0027
	SMOTE	77.0	14.0	0.0266	0.0253
	BorSMOTE	90.5	14.5	0.0150	0.0157
	ADASYN	78.5	26.5	0.1114	0.0962
	MDO	82.5	8.5	0.0071	0.0088
	MC-BRO	94.5	10.5	0.0060	0.0076
G	Base	90.0	1.0	4.882E-4	0.0015
	SMOTE	88.5	16.5	0.0224	0.0219
	BorSMOTE	82.0	9.0	0.0081	0.0097
	ADASYN	73.0	18.0	0.0574	0.0484
	MDO	82.0	9.0	0.0081	0.0097
	MC-BRO	74.5	16.5	0.0438	0.0392

data sets 1, 4, 11, and 13, respectively. This may be because these methods may cause overlaps and be more sensitive to various imbalanced data than other methods.

Figure 11 shows the top 2 results on each metric. Unexpectedly, each pair of metrics are consists of GLOS and the best one among other methods. We also mark the corresponding methods at the top of each metric in this figure. As we can see, GLOS results are better than other methods in five out of seven metrics. In the rest two metrics, such as P_{minave} and P_{min} , though the results of GLOS are worse than another, the difference can be neglected. These results suggest that GLOS is more effective than other methods for handling multi-class imbalanced classification problems. GLOS considers the overall data distribution information into account and utilizes different oversampling methods to generate instances for the minority class instances with different inherent characteristics. By doing so, the synthetic instances may strengthen the representativeness of the

minority classes without severely damaging the majority classes' accuracy.

In order to examine the robustness of GLOS in different imbalanced situations, we selected *Segment* and *Texture* data sets and modified them to different imbalanced cases. Figures 12 and 13 show the results of different methods on all metrics in three kinds of imbalanced cases, respectively. The normal case has multi-minority and multi-majority. The Multi-Minority case has multi-minority and only one majority class, while the Multi-Majority one has multi-majority and one minority class. As we can see from Fig. 12, GLOS does well in G , G_{minave} , G_{min} , P_{ave} and CBA . Moreover, in these three cases, GLOS obtained the top2 results in all metrics except for P_{minave} and P_{min} , especially it obtained the best results in G , G_{minave} and G_{min} in all cases. As shown in Fig. 13, GLOS obtained the best results on all metrics among all imbalanced cases. Moreover, the differences between top2 results in Multi-Minority and Multi-Majority cases are more significant than in the normal one, which indicates that GLOS is more effective in specific imbalanced data sets. The reasons may be that the distribution of the normal case is more complicated than other cases, and the relations between classes may not be clear at all. However, the imbalance in the Multi-Minority case can be alleviated by oversampling easier. Moreover, in the Multi-Majority case, by generating more instances, the representativeness of the minority classes could be strengthened, though these instances may damage the majority classes' performance. Furthermore, GLOS obtained the best results in G , G_{minave} , G_{min} and the best overall performance CBA , which shows that GLOS is robust and can be applied to multi-class imbalanced problems generally.

Figure 14 shows the mean results of all methods on each metric in data sets 16–21. As we can see, for each metric, the data processed by various oversampling methods have better results than the base one, and the improvements in G , G_{minave} and CBA are more obvious than other metrics. MDO has the same G_{min} with the base one, and the other methods are better than the base one, and GLOS obtains the best result. Though MDO identified the most minority class instances in the normal and Multi-Majority cases, the overlap between different minority classes in

Table 3 Summary of the Wilcoxon test of all methods on Recall

		Base	SMOTE	BorSMOTE	ADASYN	MDO	MC-BRO
GLOS	G_{min}	•	•	•	•		◦
	G_{minave}	•	•	•		•	•
	G	•	•	•	◦	•	•

Note: •(◦) indicates that the method in the row improves the method of the column with confident level $\alpha = 0.95(0.90)$

Table 4 The Wilcoxon test for GLOS and comparative algorithms on Precision

VS		R^+	R^-	Exact P -value	Asymp- totic P -value
P_{min}	Base	64.5	26.5	0.5955	0.1730
	SMOTE	53.5	37.5	≥ 0.2	0.5525
	BorSMOTE	57.5	33.5	≥ 0.2	0.3824
	ADASYN	67.5	37.5	≥ 0.2	0.3267
	MDO	86.0	19.0	0.03528	0.0328
	MC-BRO	71.5	19.5	0.0742	0.0640
P_{minave}	Base	80.5	24.5	0.0845	0.0736
	SMOTE	56.0	35.0	≥ 0.2	0.4420
	BorSMOTE	54.5	50.5	≥ 0.2	0.8753
	ADASYN	66.0	25.0	0.1677	0.1422
	MDO	78.0	13.0	0.0215	0.0211
	MC-BRO	72.5	18.5	0.0628	0.0546
P_{ave}	Base	89.0	16.0	0.0203	0.0202
	SMOTE	66.0	25.0	0.1677	0.1422
	BorSMOTE	80.0	25.0	0.0906	0.0788
	ADASYN	88.0	17.0	0.0245	0.0238
	MDO	96.0	9.0	0.0040	0.0057
	MC-BRO	83.5	21.5	0.0537	0.0480

the Multi-Minority case makes the most minority class instances hard to identify. Although GLOS, BorSMOTE and SMOTE have good performances in P_{min} and P_{minave} , BorSMOTE is worse than SMOTE in G and CBA . The reasons may be that BorSMOTE is more sensitive to the change of data imbalance than SMOTE, and focusing more on borderline instances leads the synthetic data to deteriorate the majority classes. ADASYN and MDO also have the same disadvantage, magnified clearly in the Multi-Minority and Multi-Majority cases. In this figure, SMOTE

has achieved the second best results after GLOS. It tells us that SMOTE can be an alternative to handle imbalanced data with less data complexity, such as the Multi-Minority or Multi-Majority cases. GLOS is better than other methods on all metrics except for P_{minave} and P_{min} . At the same time, GLOS has achieved the second best results on P_{minave} and P_{min} , and the recall of all components has a significant difference between GLOS and other methods. It confirms that GLOS can be applied to imbalanced data generally. Importantly, it can also improve the minority class performances without damaging the majority classes severely.

In this paper, we use the Wilcoxon ranking test to compare all methods and conduct the statistical test of significance with confidence levels $\alpha = 0.9$ and $\alpha = 0.95$, respectively. Table 2 shows the detailed statistical results that compared GLOS with other methods. The smaller the p -value is, the more likely GLOS is significantly better than the corresponding method. From Table 2, we can see that the difference on G between GLOS and the base is more significant than others, and it also improves G_{minave} significantly than other methods except for MDO. It indicates that GLOS can identify more classes than other methods, and the minority class instances can be identified without severely misclassifying the majority class instances into the minority classes.

From Table 3 we can see that all methods have improvements in G_{min} , G_{minave} , and G significantly compared with the base classifier. Moreover, GLOS has significant improvements in most metrics compared with other methods. However, GLOS has no significant improvement in G_{min} relative to MDO because MDO can identify more minority class instances even though it may sacrifice the majority class performance. Though GLOS and ADASYN have no significant difference in G_{minave} , GLOS is still better than ADASYN as GLOS obtained better results in other aspects.

Table 5 Summary of the Wilcoxon test of all methods on Precision

		Base	SMOTE	BorSMOTE	ADASYN	MDO	MC-BRO
GLOS	P_{min}					•	◦
	P_{minave}	◦				•	◦
	P_{ave}	•		◦	◦	•	•

Table 6 Summary of the Wilcoxon test for GLOS and comparative algorithms on all metrics

		Base	SMOTE	BorSMOTE	ADASYN	MDO	MC-BRO
GLOS	P_{min}						
	P_{minave}						
	P_{ave}	◦		◦	◦		•
	G_{min}	◦	◦	◦	◦	◦	◦
	G_{minave}	◦	◦	◦		◦	◦
	G	◦		◦	◦	◦	•
	CBA	◦		◦	◦	◦	•

Table 4 shows the detailed statistical results that compared GLOS with other methods. The smaller the p -value is, the more significantly GLOS is better than the corresponding method. It shows that GLOS can improve P_{ave} significantly than other methods at level $\alpha = 0.95$. It also performs better than MC-BRO and MDO in P_{minave} and P_{min} with confident level $\alpha = 0.9$. From Table 5 we can see that almost methods except GLOS have no significant difference in P_{min} , P_{minave} , and P_{ave} compared with the base results. Moreover, BorSMOTE has significant improvement in P_{minave} compared to the base cases, ADASYN, and MDO at level $\alpha = 0.95$. Though GLOS has no significant difference with MDO in G_{min} , GLOS improved P_{min} significantly than MDO. In a word, GLOS can obtain better results in each component than other methods.

Table 6 shows the results of the Wilcoxon test that compared GLOS to all methods on data sets (16–21), from it we can see that GLOS improved P_{ave} , G_{min} , G_{minave} , G and CBA than almost other methods significantly at level $\alpha = 0.9$. And all methods have no significant difference in P_{min} and P_{minave} . Moreover, it also inspired us that SMOTE can make unexpected performance in the multi-minority or multi-majority imbalanced cases. Because SMOTE has significant improvements than other methods with confident level $\alpha = 0.95$ in P_{ave} , G_{minave} , G and CBA . In a word, GLOS can obtain better results in most imbalanced cases compared with other methods. Importantly, it makes great improvements in G , G_{minave} and G_{min} significantly, which means that GLOS can identify the minority classes well in most imbalanced cases.

5 Conclusion

Most oversampling methods use a single approach to synthesize instances for hard-to-learn minority class instances to reduce the class imbalance. However, they are not always suitable for all instances to be synthesized, which limits the effectiveness of some synthesized ones to a certain extent and even deteriorates the classification performance of the original data. The proposed GLOS utilizes different synthesis strategies to purposefully generate instances according to the intrinsic characteristics of different instances. By doing so, synthetic instances' diversity and quality are improved. Experimental results show that GLOS significantly outperforms other methods, especially in improving minority class recall. Importantly, it inspires us to gain insight into different data characteristics and address the impact of these characteristics on classification specifically, which can help to solve the imbalance problem better.

However, many additional factors affecting the classification complexity are worth exploring in the future. Moreover, our method involves two parameters in computing DID

and instances separation, and these parameters are closely related to the results. We have given the recommended values of these parameters according to experiment results, but it is necessary to reduce the dependence of GLOS on these parameters. Future research will focus on this problem, and we intend to obtain suitable parameters according to data complexity adaptively.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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